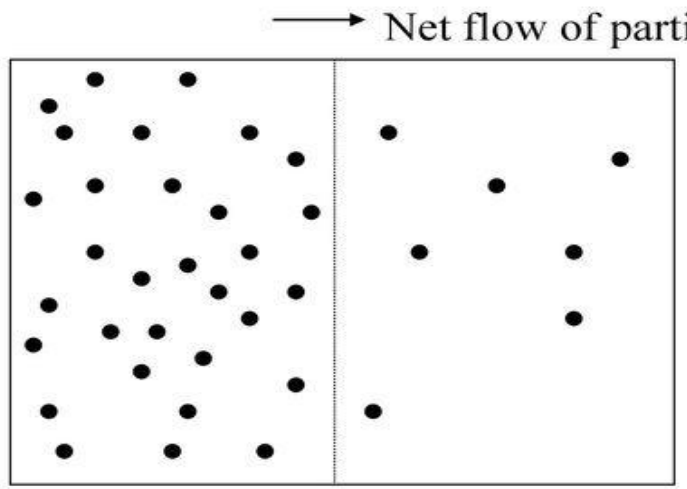
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4.4 Diffusion of Carriers

4.4.1 Diffusion Processes

Two basic processes of current conduction:

- *Drift* in an electric field
- *Diffusion* due to a concentration gradient



Diffusion is the result of the *random motion* the individual particles!

Ohm's Law

$$J = \sigma E$$

J is the current density, coulombs/m^2
For the Earth's atmosphere, J is
1.5×10^{-12} A m^{-2}

Here the symbol σ is the conductivity with units of S/m, Siemens/meter, of Mhos

Conductivity is a measure of the ease which charge moves through a given medium.

Ohm's Law may also be expressed in simple "circuit" form, V=IR where V is The voltage, I is the current and R is the resistance (Ohms)

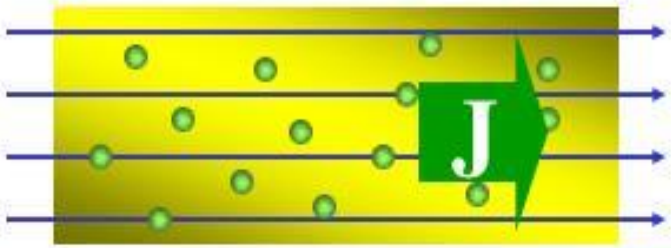


Current And Resistance

Current Density and Drift Velocity

- *Perfect conductors* carry charge instantaneously from here to there
- *Perfect insulators* carry no charge from here to there, ever

- Real substances *always* have some density n of charges q that can move, however slowly
  - Usually electrons
- When you turn on an electric field, the charges start to move with average velocity $\mathbf{v}_d$
  - Called the *drift velocity*



Why did I draw **J** to the right?

- There is a *current density* **J** associated with this motion of charges
  - Current density represents a flow of charge
- Note: **J** tends to be in the direction of **E**, even when $\mathbf{v}_d$ isn't

$$\mathbf{J} = nq\mathbf{v}_d$$

❖ Carrier Drift

- Electron and holes will move under the influence of an applied electric field since the field exert a force on charge carriers (electrons and holes).

$$F = qE$$

- These movements result a current of I_d;

$$I_d = nqV_dA$$

I_d : drift current

V_d : drift velocity of charge carrier

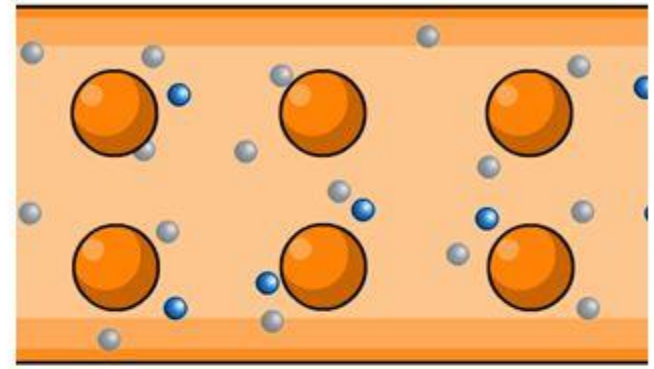
A : area of the semiconductor

n : number of charge carriers per unit volume

q : charge of the electron

Current and drift velocity

Current is a flow of charge. Electrical devices activate almost instantly once they are supplied with power, however the electrons actually move around a circuit quite slowly. Their velocity is called **drift velocity**.



Current and drift velocity are linked by the following equation:

$$I = nAve$$

I = current (amps)

n = charged particles per unit volume

A = cross-sectional area (m^2)

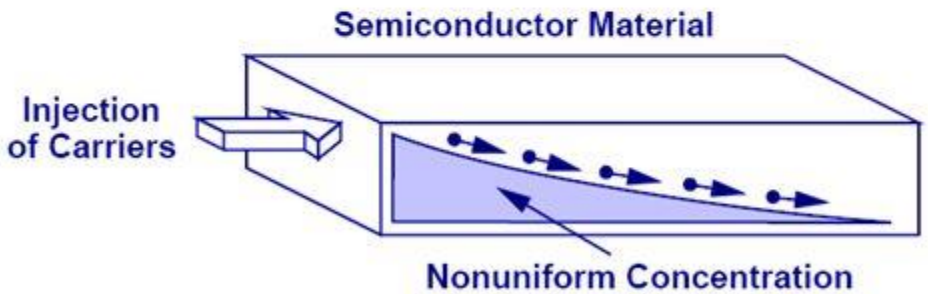
v = drift velocity (m/s)

e = charge on an electron (1.6×10^{-19}C)

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Carrier Diffusion

- Due to thermally induced random motion, mobile particles tend to move from a region of high concentration to a region of low concentration.
  - Analogy: ink droplet in water
- Current flow due to mobile charge diffusion is proportional to the carrier concentration gradient.
  - The proportionality constant is the ***diffusion constant***.



$$J_p = -qD_p \frac{dp}{dx}$$

Notation:

$D_p \equiv$ hole diffusion constant (cm^2/s)

$D_n \equiv$ electron diffusion constant (cm^2/s)

Diffusion Current



- Diffusion current within a semiconductor consists of hole and electron components:

$$J_{p,diff} = -qD_p \frac{dp}{dx} \qquad J_{n,diff} = qD_n \frac{dn}{dx}$$

$$J_{tot,diff} = q\left(D_n \frac{dn}{dx} - D_p \frac{dp}{dx}\right)$$

- The total current flowing in a semiconductor is the sum of drift current and diffusion current:

$$\boxed{J_{tot} = J_{p,drift} + J_{n,drift} + J_{p,diff} + J_{n,diff}}$$

Diffusion Current

The total current flowing in a semiconductor is the sum of drift current and diffusion current:

$$J_{tot} = J_{p,drift} + J_{n,drift} + J_{p,diff} + J_{n,diff}$$

- The characteristic constants for drift and diffusion are related by the Einstein's Relation:

$$\frac{D}{\mu} = \frac{kT}{q}$$

- $\frac{kT}{q} \cong 26\text{mV}$ at room temperature (300K)

This is often referred to as the **"thermal voltage"**.

THERMOELECTRIC EFFECT OF CURRENT

The phenomenon of production of electricity with the help of heat is called thermoelectricity and this effect is called thermoelectric effect.

Or

The phenomenon of generation of an electric current in a thermocouple by keeping its junctions at different temperatures is called thermoelectric effect.

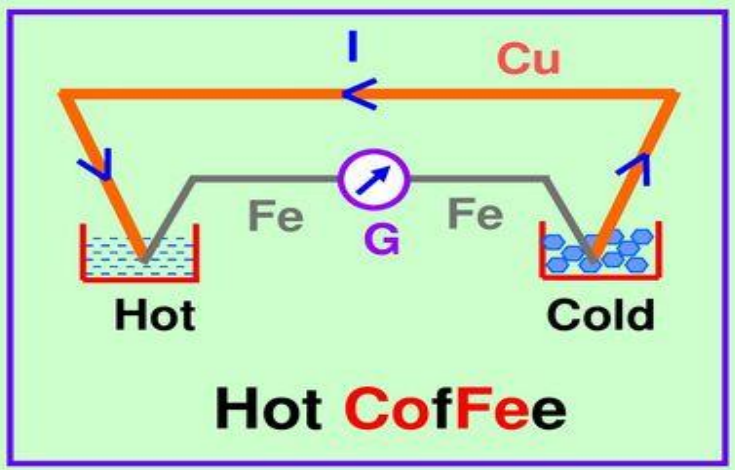
An arrangement of two wires of different materials (conductors) joined at their ends to form two junctions is called thermocouple.

Seebeck Effect:

$$E = \alpha t + \tfrac{1}{2} \beta t^2$$

where α and β are thermoelectric constants

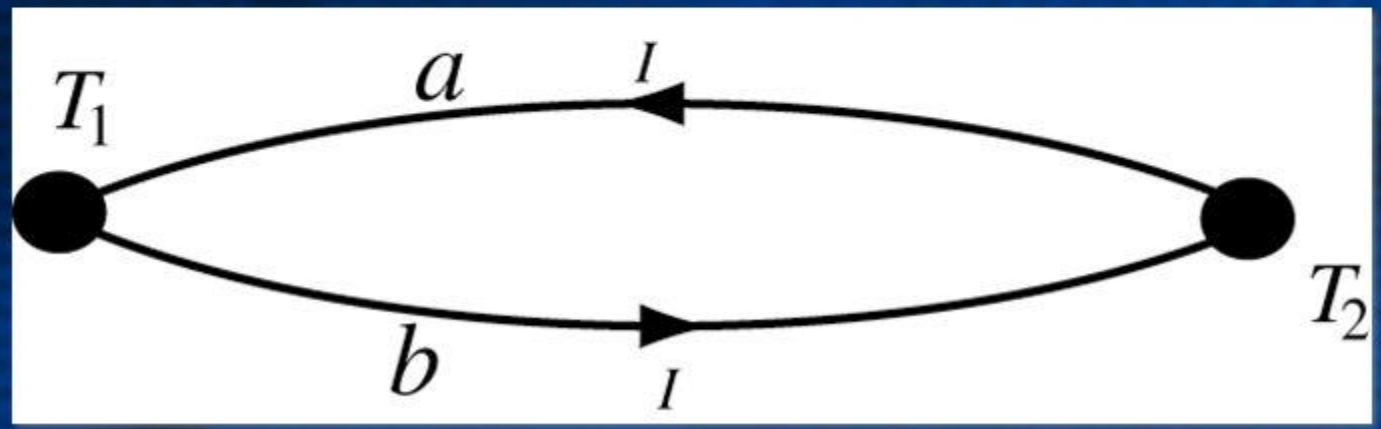
The current produced in this effect is called thermoelectric current and the emf is called thermoelectric emf.



Thermoelectric emf is usually in the order of μV.

The Seebeck effect

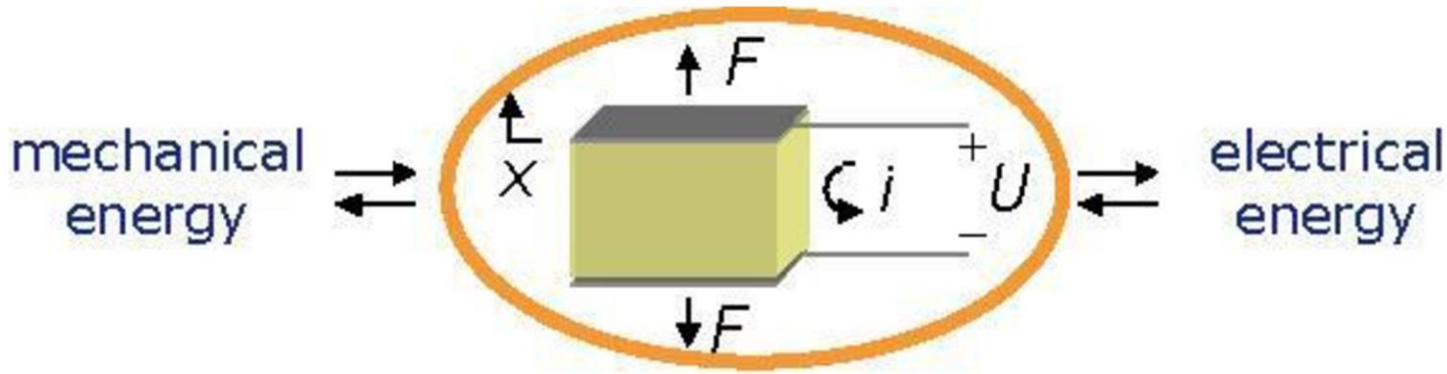
- If both ends of the two conductors are connected and a temperature difference is maintained between the two junctions, a thermoelectric current will flow through the closed circuit (generation mode)



Piezoelectric effect

- The appearance of electric potential across face of crystal when it is subjected to mechanical pressure

- Conversely, when an electric an electric field is applied on one face of the crystal it undergoes mechanical distortion



Photovoltaic Effect

- The creation of a voltage (or a corresponding electric current) in a material upon exposure to electro magnetic radiation is known as Photo-voltaic effect.
- Certain semiconductors materials absorb certain wavelengths
- An Example of a semiconductor is crystalline silicon.

